(a)

Tooth detector
class ‘tooth’; boxes kept if score ≥ 0.3

Numbering model
32 FDI classes; class-agnostic NMS, best box per code

Number assignment
each detector box, by score, takes the unused numbering box of highest IoU if IoU ≥ 0.5; otherwise the tooth stays unassigned

Consistency check (no number is forced)
review if the code is duplicated, confidence < 0.5, or the box side / arch disagrees with the quadrant

Mask matching
Mask R-CNN instance ↔ detector box, one-to-one, IoU ≥ 0.5

(b)

